

2.2 Linear and affine transformations

Almost every statistic in a linear model is a linear function of the response: least squares coefficients, fitted values, residuals, contrasts. Their means and covariance matrices follow from one theorem.

2.2.1 The transformation rules

Theorem 2.2.1 — Moments of affine transformations

Let $\mathbf{Y}$ be a $p \times 1$ random vector with finite second moments, $E(\mathbf{Y}) = \boldsymbol{\mu}$ and $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$. Let $\mathbf{A}$ be $k \times p$, $\mathbf{B}$ be $m \times p$, and $\mathbf{b}$ be $k \times 1$, all constant. Then

- (a) $E(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$;
- (b) $\text{Cov}(\mathbf{AY} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$;
- (c) $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$.

More generally, if $\mathbf{U}$ and $\mathbf{V}$ have finite second moments and $\mathbf{A}$, $\mathbf{B}$, $\mathbf{b}$, $\mathbf{c}$ are constants of conformable sizes, then $\text{Cov}(\mathbf{AU} + \mathbf{b}, \mathbf{BV} + \mathbf{c}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^\top$.

Proof.

- (a) is Proposition 2.1.1(a). For the general statement, the centred vectors are $\mathbf{AU} + \mathbf{b} - E(\mathbf{AU} + \mathbf{b}) = \mathbf{A}(\mathbf{U} - E\mathbf{U})$ and similarly $\mathbf{B}(\mathbf{V} - E\mathbf{V})$. Hence

$$\text{Cov}(\mathbf{AU} + \mathbf{b}, \mathbf{BV} + \mathbf{c}) = E[\mathbf{A}(\mathbf{U} - E\mathbf{U})(\mathbf{V} - E\mathbf{V})^\top \mathbf{B}^\top] = \mathbf{A} E[(\mathbf{U} - E\mathbf{U})(\mathbf{V} - E\mathbf{V})^\top] \mathbf{B}^\top,$$

again by Proposition 2.1.1(a). Parts (c) and (b) are the cases $\mathbf{U} = \mathbf{V} = \mathbf{Y}$, and $\mathbf{B} = \mathbf{A}$ in (b). ■

For a single linear combination, $\mathbf{A} = \mathbf{a}^\top$ is a row vector, and (b) and (c) give

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \sum_{i=1}^p \sum_{j=1}^p a_i a_j \sigma_{ij}, \quad \text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{b}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{b}. \quad (2.2.1)$$

Every variance and covariance enters $\text{Var}(\mathbf{a}^\top \mathbf{Y})$, weighted by the products of the coefficients. Conversely, $\boldsymbol{\Sigma}$ is determined by the variances of linear combinations, because $\sigma_{ij} = \frac{1}{2}[\text{Var}(Y_i + Y_j) - \text{Var}(Y_i) - \text{Var}(Y_j)]$.

Proposition 2.2.2 — Covariance of sums

Let $\mathbf{U}, \mathbf{U}_2, \mathbf{V}$ have finite second moments, with $\mathbf{U}$ and $\mathbf{U}_2$ of the same size in (a), and $\mathbf{U}$ and $\mathbf{V}$ of the same size in (b). Then

- (a) $\text{Cov}(\mathbf{U} + \mathbf{U}_2, \mathbf{V}) = \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{U}_2, \mathbf{V})$;
- (b) $\text{Cov}(\mathbf{U} + \mathbf{V}) = \text{Cov}(\mathbf{U}) + \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{V}, \mathbf{U}) + \text{Cov}(\mathbf{V})$;
- (c) if $\mathbf{Y}_1, \dots, \mathbf{Y}_n$ are pairwise uncorrelated $p \times 1$ vectors and $c_1, \dots, c_n$ are constants, then $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i c_i^2 \text{Cov}(\mathbf{Y}_i)$. In particular, if every $\text{Cov}(\mathbf{Y}_i) = \mathbf{\Sigma}$, then $\text{Cov}(\bar{\mathbf{Y}}) = \mathbf{\Sigma}/n$.

Proof.

- (a) Centre each vector and use linearity of the outer product and of E. (b) Apply (a) in each argument; note that the two cross terms are transposes of each other, not equal.
- (b) By (a) and (b), applied repeatedly, $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i \sum_j c_i c_j \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j)$, using $\text{Cov}(c\mathbf{U}, \mathbf{V}) = c \text{Cov}(\mathbf{U}, \mathbf{V})$ from [Theorem 2.2.1](#). The terms with $i \neq j$ vanish. ■

Example. (Centring a sample)

Let $\mathbf{Y}$ be $n \times 1$ with $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$, and write $\mathbf{J} = \mathbf{1}\mathbf{1}^\top$. The mean is $\bar{Y} = n^{-1}\mathbf{1}^\top \mathbf{Y}$, and the vector of deviations is $\mathbf{d} = (\mathbf{I} - n^{-1}\mathbf{J})\mathbf{Y}$. The matrix $\mathbf{I} - n^{-1}\mathbf{J}$ is symmetric and idempotent, so [Theorem 2.2.1](#) gives

$$\text{Cov}(\mathbf{d}) = \sigma^2(\mathbf{I} - n^{-1}\mathbf{J}), \quad \text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1}\mathbf{J}) = \mathbf{0}^\top.$$

The deviations have variances $\sigma^2(1 - 1/n)$ and are negatively correlated, with correlation $-1/(n-1)$ between each pair, because they are forced to sum to zero. And each deviation is uncorrelated with the mean. Under normality, uncorrelated becomes independent (Example (Sample mean and deviations)), which is the first half of Corollary 4.4.5.

2.2.2 Covariance matrices are nonnegative definite

A variance cannot be negative, and by [Eq. 2.2.1](#) this simple fact constrains the whole matrix. Recall from Definition 1.7.1 that a symmetric $\mathbf{\Sigma}$ is nonnegative definite if $\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a} \geq 0$ for all $\mathbf{a}$, and positive definite if in addition $\mathbf{a}^\top \mathbf{\Sigma} \mathbf{a} > 0$ for $\mathbf{a} \neq \mathbf{0}$.

Theorem 2.2.3 — Covariance matrices

Let $\mathbf{Y}$ have finite second moments, mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{\Sigma}$.

- (a) $\mathbf{\Sigma}$ is nonnegative definite.
- (b) Conversely, every $p \times p$ nonnegative definite matrix is the covariance matrix of some random vector.
- (c) For a constant $\mathbf{a}$, $\mathbf{\Sigma} \mathbf{a} = \mathbf{0}$ iff $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu}$ with probability one.
- (d) $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\mathbf{\Sigma})) = 1$. If $\mathcal{S}$ is any subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$, then $\mathcal{C}(\mathbf{\Sigma}) \subseteq \mathcal{S}$.

In particular, $\mathbf{\Sigma}$ is singular iff some nontrivial linear combination of $\mathbf{Y}$ is constant with

probability one. Then $\mathbf{Y}$ lies, with probability one, in the affine subspace $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$ of dimension $\text{rank}(\boldsymbol{\Sigma}) < p$, and this is the smallest such flat.

Proof.

- (a) $\boldsymbol{\Sigma}$ is symmetric, and $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \text{Var}(\mathbf{a}^\top \mathbf{Y}) \geq 0$ by [Eq. 2.2.1](#).
- (b) Let $\boldsymbol{\Sigma}$ be nonnegative definite and let $\boldsymbol{\Sigma}^{1/2}$ be its symmetric nonnegative definite square root (Theorem 1.7.6). Let $\mathbf{Z}$ have independent components with mean 0 and variance 1, for instance independent standard normal variables, so that $\text{Cov}(\mathbf{Z}) = \mathbf{I}$. Then $\text{Cov}(\boldsymbol{\Sigma}^{1/2} \mathbf{Z}) = \boldsymbol{\Sigma}^{1/2} \mathbf{I} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$.
- (c) If $\boldsymbol{\Sigma} \mathbf{a} = \mathbf{0}$, the random variable $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu})$ has mean zero and variance $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = 0$, so it is zero with probability one. Conversely, if $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, then $\boldsymbol{\Sigma} \mathbf{a} = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top \mathbf{a}] = \mathbf{0}$.
- (d) Let $\mathbf{a}_1, \dots, \mathbf{a}_s$ be a basis of $\mathcal{N}(\boldsymbol{\Sigma})$. By (c), each event $\{\mathbf{a}_j^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0\}$ has probability one, and so does their intersection. On that intersection $\mathbf{Y} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{N}(\boldsymbol{\Sigma})$. Because $\boldsymbol{\Sigma}$ is symmetric, $\mathcal{N}(\boldsymbol{\Sigma})$ is the orthogonal complement of $\mathcal{C}(\boldsymbol{\Sigma})$, and so $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$. Now let $\mathcal{S}$ be a subspace with $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$. For every $\mathbf{a} \in \mathcal{S}^\perp$ we have $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$ with probability one, hence $\mathbf{a} \in \mathcal{N}(\boldsymbol{\Sigma}) = \mathcal{C}(\boldsymbol{\Sigma})^\perp$ by (c). So $\mathcal{S}^\perp \subseteq \mathcal{C}(\boldsymbol{\Sigma})^\perp$, and taking orthogonal complements gives $\mathcal{C}(\boldsymbol{\Sigma}) \subseteq \mathcal{S}$. The final statements follow, with $\dim \mathcal{C}(\boldsymbol{\Sigma}) = \text{rank}(\boldsymbol{\Sigma})$. For minimality among all flats, suppose $\Pr(\mathbf{Y} \in \mathbf{c} + \mathcal{S}) = 1$ for a vector $\mathbf{c}$ and a subspace $\mathcal{S}$. Every $\mathbf{a} \in \mathcal{S}^\perp$ has $\mathbf{a}^\top (\mathbf{Y} - \mathbf{c}) = 0$ with probability one, and taking expectations gives $\mathbf{a}^\top (\boldsymbol{\mu} - \mathbf{c}) = 0$. Hence $\boldsymbol{\mu} - \mathbf{c} \in \mathcal{S}$, so $\mathbf{c} + \mathcal{S} = \boldsymbol{\mu} + \mathcal{S}$, and the subspace case applies. ■

A covariance matrix is therefore positive definite exactly when no linear combination of the components is degenerate. Singular covariance matrices are not pathological. They arise whenever the components satisfy an exact linear constraint, as with the deviations of [Example \(Centring a sample\)](#), whose covariance matrix has rank $n - 1$ because $\mathbf{1}^\top \mathbf{d} = 0$. The residual vector of a regression is another case: it is orthogonal to every column of $\mathbf{X}$, and when $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ its covariance matrix has rank $n - \text{rank}(\mathbf{X})$ ([Section 2.2, Chapter 6](#)).

Example. (Multinomial counts)

In m independent trials, each outcome falls into one of k categories with probabilities $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k)^\top$, where $\sum_j \pi_j = 1$. Let $\mathbf{Z}_t$ be the indicator vector of trial t , with a single 1 in the position of its category. Then $\mathbb{E}(\mathbf{Z}_t) = \boldsymbol{\pi}$, and $\mathbf{Z}_t \mathbf{Z}_t^\top = \text{diag}(\mathbf{Z}_t)$ because the indicator has only one nonzero entry. So $\text{Cov}(\mathbf{Z}_t) = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top$. The count vector $\mathbf{Y} = \sum_t \mathbf{Z}_t$ is a sum of independent vectors, and [Proposition 2.2.2\(c\)](#) gives

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y}) = m(\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top).$$

Because $\boldsymbol{\Sigma} \mathbf{1} = m(\boldsymbol{\pi} - \boldsymbol{\pi} \boldsymbol{\pi}^\top \mathbf{1}) = \mathbf{0}$, the matrix is singular, and [Theorem 2.2.3\(c\)](#) recovers the constraint $\mathbf{1}^\top \mathbf{Y} = m$. For $m = 20$ and $\boldsymbol{\pi} = (0.5, 0.3, 0.2)^\top$ the eigenvalues are 7.762, 4.638 and 0. All 5000 simulated count vectors lie in the plane $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma}) = \{\mathbf{y} : \mathbf{1}^\top \mathbf{y} = m\}$.

Linear maps cannot increase the rank of a covariance matrix, since $\text{rank}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top) \leq \text{rank}(\boldsymbol{\Sigma})$ ([Exercise \(B2\)](#)). They can, however, discard degenerate directions. If $\mathbf{Y} = (Z, Z)^\top$ with $\text{Var}(Z) = 1$, then $\boldsymbol{\Sigma}$ has all entries equal to 1 and is singular, while $\mathbf{A} = (1, 0)$ gives $\text{Var}(\mathbf{A} \mathbf{Y}) = 1 > 0$. By

Theorem 2.2.1, the covariance matrix of $\mathbf{AY}$ is $\mathbf{A}\Sigma\mathbf{A}^\top$. If Σ is positive definite and $\mathbf{A}$ has full row rank $k \leq p$, then for $\mathbf{b} \neq \mathbf{0}$ the vector $\mathbf{A}^\top\mathbf{b}$ is nonzero and $\mathbf{b}^\top\mathbf{A}\Sigma\mathbf{A}^\top\mathbf{b} > 0$. So $\mathbf{A}\Sigma\mathbf{A}^\top$ is positive definite. If $k > p$, then $\mathbf{A}\Sigma\mathbf{A}^\top$ has rank at most p and is singular. A vector of more than p linear combinations of p variables always satisfies a linear constraint.

2.2.3 Standardization and Mahalanobis distance

A scalar with mean μ and variance $\sigma^2 > 0$ is standardized by $(Y - \mu)/\sigma$. The vector version must remove correlation as well as scale.

Proposition 2.2.4 — Whitening

Let $\Sigma = \text{Cov}(\mathbf{Y})$ be positive definite, with $E(\mathbf{Y}) = \mu$.

- (a) $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{Y} - \mu)$ has $E(\mathbf{Z}) = \mathbf{0}$ and $\text{Cov}(\mathbf{Z}) = \mathbf{I}_p$.
- (b) A $p \times p$ matrix $\mathbf{W}$ satisfies $\mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{I}_p$ iff $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$ for some orthogonal $\mathbf{Q}$. In particular, if $\Sigma = \mathbf{L}\mathbf{L}^\top$ is the Cholesky factorization, $\mathbf{W} = \mathbf{L}^{-1}$ is such a matrix.
- (c) For every $\mathbf{W}$ as in (b), $\|\mathbf{W}(\mathbf{Y} - \mu)\|^2 = (\mathbf{Y} - \mu)^\top \Sigma^{-1}(\mathbf{Y} - \mu)$.

Proof.

- (a) By **Theorem 2.2.1**, $\text{Cov}(\mathbf{Z}) = \Sigma^{-1/2}\Sigma\Sigma^{-1/2} = \mathbf{I}$, since $\Sigma^{-1/2}$ is symmetric and commutes with Σ . (b) If $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$, then $\mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$. Conversely, if $\mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{I}$, put $\mathbf{Q} = \mathbf{W}\Sigma^{1/2}$. Then $\mathbf{Q}\mathbf{Q}^\top = \mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{I}$, so the square matrix $\mathbf{Q}$ is orthogonal, and $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$. For the Cholesky factor, $\mathbf{L}^{-1}\Sigma(\mathbf{L}^\top)^{-1} = \mathbf{L}^{-1}\mathbf{L}\mathbf{L}^\top(\mathbf{L}^\top)^{-1} = \mathbf{I}$. (c) $\mathbf{W}^\top\mathbf{W} = \Sigma^{-1/2}\mathbf{Q}^\top\mathbf{Q}\Sigma^{-1/2} = \Sigma^{-1}$.

■

The quantity in (c),

$$\Delta^2(\mathbf{Y}) = (\mathbf{Y} - \mu)^\top \Sigma^{-1}(\mathbf{Y} - \mu), \quad (2.2.2)$$

is the squared **Mahalanobis distance** of $\mathbf{Y}$ from μ . It measures distance in units of the covariance structure. Its level sets are ellipsoids with axes along the eigenvectors of Σ , with half-lengths proportional to the square roots of the eigenvalues. Whitening maps these ellipsoids to spheres. Part (b) says that whitening is unique only up to a rotation or reflection. Different choices give different coordinates, but the same distances.

Example. (Two whitenings)

Take $\mu = (3, 1)^\top$ and $\Sigma = \begin{pmatrix} 4 & 2.4 \\ 2.4 & 2.25 \end{pmatrix}$, so the standard deviations are 2 and 1.5 and the correlation is 0.8. The eigenvalues are 5.680 and 0.570. The symmetric and Cholesky whitening matrices are

$$\Sigma^{-1/2} = \begin{pmatrix} 0.717 & -0.425 \\ -0.425 & 1.027 \end{pmatrix}, \quad \mathbf{L}^{-1} = \begin{pmatrix} 0.500 & 0 \\ -0.667 & 1.111 \end{pmatrix}.$$

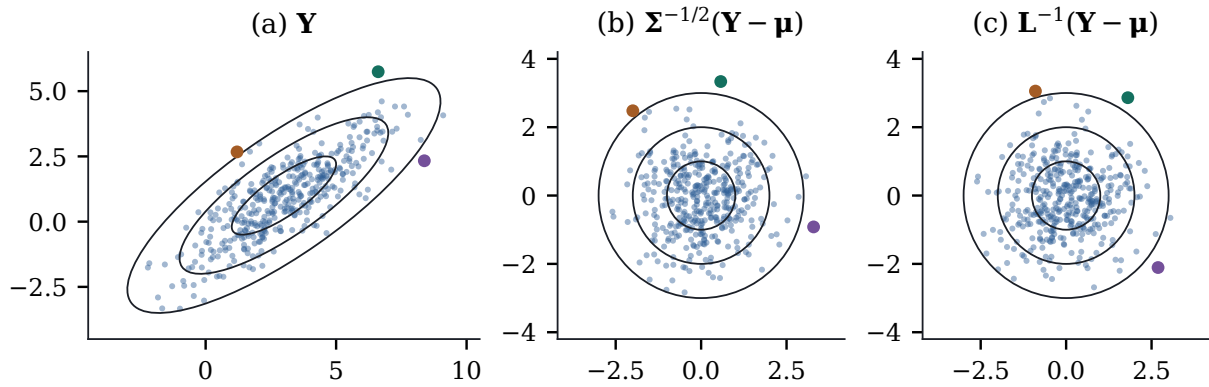


Figure 2.2.1: Whitening a correlated sample. (a) Draws with $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ and the Mahalanobis contours $\Delta = 1, 2, 3$. (b) The symmetric whitening $\boldsymbol{\Sigma}^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$. (c) The Cholesky whitening $\mathbf{L}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$. The contours become circles in both, and the three coloured points keep their distance from the origin. Panels (b) and (c) differ by a rotation (Proposition 2.2.4).

The Cholesky version standardizes Y_1 alone and then removes from Y_2 its linear dependence on Y_1 , in the spirit of Gram–Schmidt. The symmetric version treats both coordinates alike and moves points as little as possible (Exercise (B3)). Figure 2.2.1 applies both to 400 simulated points. The two whitened clouds differ by a rotation through 22.5° , and every point keeps its Mahalanobis distance.

```
import numpy as np
rng = np.random.default_rng(2)

mu = np.array([3.0, 1.0])
Sigma = np.array([[4.0, 2.4],
                  [2.4, 2.25]])
0.8                                     # correlation 2.4 / (2 * 1.5) =

lam, U = np.linalg.eigh(Sigma)
W_sym = U @ np.diag(lam ** -0.5) @ U.T
L = np.linalg.cholesky(Sigma)
W_chol = np.linalg.inv(L)

# spectral decomposition
# Sigma^{-1/2}
# Sigma = L L', L lower triangular
# L^{-1}

n = 400
Y = mu + rng.standard_normal((n, 2)) @ L.T
do                                     # any draws with Cov = Sigma will
Z_sym = (Y - mu) @ W_sym.T
Z_chol = (Y - mu) @ W_chol.T

Q = W_chol @ np.linalg.inv(W_sym)
print("Q'Q =", np.round(Q.T @ Q, 12))
# L^{-1} Sigma^{1/2}
```

[Run this code online](#)

Whitening runs in both directions. If $\text{Cov}(\mathbf{Z}) = \mathbf{I}$, then $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$, or $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$, has mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$. This is how correlated vectors are simulated, and with $\mathbf{Z}$ standard normal it is how Chapter 3 constructs $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$. The Mahalanobis distance then appears in the exponent of the density (Theorem 3.2.2). When $\boldsymbol{\Sigma}$ is singular, no matrix whitens all of $\mathbf{Y}$, since a degenerate combination cannot be scaled to variance one. The best one can do is whiten within the support $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$ using the Moore–Penrose inverse square root (Exercise (B4)).

2.2.4 Second moments in the linear model

Here is how the rules of this section enter regression. Suppose $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ an $n \times p$ constant matrix of rank p , $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ and $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$. Adding the constant $\mathbf{X}\boldsymbol{\beta}$ changes the mean but not the covariance, so $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ and $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$. The least squares estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ is $\mathbf{A}\mathbf{Y}$ with $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$, so

$$E(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

No distributional assumption was used, and none is needed for the Gauss–Markov theorem (Theorem 7.2.1). Fitted values and residuals are also linear in $\mathbf{Y}$, and their covariance matrices are computed the same way in Chapter 5 and Chapter 6. If instead $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ with $\mathbf{V}$ positive definite, then $\mathbf{V}^{-1/2} \mathbf{Y} = \mathbf{V}^{-1/2} \mathbf{X} \boldsymbol{\beta} + \mathbf{V}^{-1/2} \boldsymbol{\varepsilon}$ is a model with $\text{Cov}(\mathbf{V}^{-1/2} \boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. Whitening the data turns correlated errors back into the standard case. This is generalized least squares, which Chapter 6 and Chapter 31 develop.

2.2.5 Exercises

2.2.5.1 A. Check your understanding

Exercise A1.

Let $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_3$. Find the covariance and correlation matrices of the successive differences $(Y_2 - Y_1, Y_3 - Y_2)^\top$, and of $(Y_2 - Y_1, Y_3 - Y_2, Y_3 - Y_1)^\top$. Which is singular, and which linear combination is degenerate?

Exercise A2.

For $p = 2$ and $\boldsymbol{\Sigma} = \mathbf{L}\mathbf{L}^\top$, show that the second component of $\mathbf{L}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ is a multiple of $Y_2 - \mu_2 - (\sigma_{12}/\sigma_{11})(Y_1 - \mu_1)$. Explain the connection with Theorem 2.5.2.

2.2.5.2 B. Practice

Exercise B1.

Let $\varepsilon_0, \dots, \varepsilon_n$ be uncorrelated with variance σ^2 , and let $Y_t = \varepsilon_t + \theta \varepsilon_{t-1}$ for $t = 1, \dots, n$. Find $\text{Cov}(\mathbf{Y})$, and show that it is positive definite for every real θ , including $\theta = \pm 1$.

Solution. $\mathbf{Y} = \mathbf{A}\boldsymbol{\varepsilon}$, where $\mathbf{A}$ is $n \times (n+1)$ with θ in column t and 1 in column $t+1$ of row t (columns indexed by $\varepsilon_0, \dots, \varepsilon_n$). So $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$ is tridiagonal, with $\sigma^2(1 + \theta^2)$ on the diagonal and $\sigma^2\theta$ next to it. The last n columns of $\mathbf{A}$ form a lower bidiagonal matrix with unit diagonal, which is nonsingular. So $\mathbf{A}$ has full row rank n , and $\mathbf{A}\mathbf{A}^\top$ is positive definite (Section 2.2) whatever the value of θ .

Exercise B2.

Show that $\mathcal{C}(\mathbf{A}\Sigma\mathbf{A}^\top) = \mathcal{C}(\mathbf{A}\Sigma)$ for every nonnegative definite Σ and conformable $\mathbf{A}$. Deduce that $\text{rank Cov}(\mathbf{A}\mathbf{Y}) \leq \text{rank Cov}(\mathbf{Y})$.

Exercise B3.

Let Σ be positive definite. Among all $\mathbf{W}$ with $\mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{I}$, show that $\mathbb{E}\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu}) - (\mathbf{Y} - \boldsymbol{\mu})\|^2$ is minimized uniquely by $\mathbf{W} = \Sigma^{-1/2}$. *Hint:* use Proposition 2.2.4(b) and show that $\text{tr}(\mathbf{Q}\Sigma^{1/2}) \leq \text{tr}(\Sigma^{1/2})$ for orthogonal $\mathbf{Q}$.

Solution. Write $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$. By Corollary 2.3.2(a) with $\mathbf{A} = (\mathbf{W} - \mathbf{I})^\top(\mathbf{W} - \mathbf{I})$,

$$\mathbb{E}\|(\mathbf{W} - \mathbf{I})(\mathbf{Y} - \boldsymbol{\mu})\|^2 = \text{tr}((\mathbf{W} - \mathbf{I})\Sigma(\mathbf{W} - \mathbf{I})^\top) = \text{tr}(\mathbf{W}\Sigma\mathbf{W}^\top) - 2\text{tr}(\mathbf{W}\Sigma) + \text{tr}(\Sigma) = p + \text{tr}(\Sigma) - 2\text{tr}(\mathbf{Q}\Sigma^{1/2}).$$

Let $\Sigma^{1/2} = \mathbf{\Gamma}\mathbf{\Lambda}^{1/2}\mathbf{\Gamma}^\top$ and $\mathbf{G} = \mathbf{\Gamma}^\top\mathbf{Q}\mathbf{\Gamma}$, which is orthogonal, so each $|g_{ii}| \leq 1$. Then $\text{tr}(\mathbf{Q}\Sigma^{1/2}) = \text{tr}(\mathbf{G}\mathbf{\Lambda}^{1/2}) = \sum_i g_{ii}\lambda_i^{1/2} \leq \sum_i \lambda_i^{1/2}$. Since every $\lambda_i > 0$, equality forces every $g_{ii} = 1$. An orthogonal matrix with unit diagonal is $\mathbf{I}$, because each column has norm one. So $\mathbf{G} = \mathbf{I}$, which means $\mathbf{Q} = \mathbf{I}$.

Exercise B4.

Let $\Sigma = \text{Cov}(\mathbf{Y})$ have rank r , with spectral decomposition $\Sigma = \mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top$, where $\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r = \mathbf{I}_r$ and $\mathbf{\Lambda}_r$ holds the positive eigenvalues. Show that $\mathbf{Z} = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top(\mathbf{Y} - \boldsymbol{\mu})$ has $\text{Cov}(\mathbf{Z}) = \mathbf{I}_r$ and that $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{\Gamma}_r\mathbf{\Lambda}_r^{1/2}\mathbf{Z}$ with probability one. Show that $(\mathbf{Y} - \boldsymbol{\mu})^\top\Sigma^+(\mathbf{Y} - \boldsymbol{\mu}) = \|\mathbf{Z}\|^2$, where Σ^+ is the Moore–Penrose inverse, and that its expectation is r .

Solution. By Theorem 2.2.1, $\text{Cov}(\mathbf{Z}) = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r^{-1/2} = \mathbf{I}_r$. By Theorem 2.2.3(d), $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma) = \mathcal{C}(\mathbf{\Gamma}_r)$ with probability one, and on that event $\mathbf{Y} - \boldsymbol{\mu} = \mathbf{\Gamma}_r\mathbf{\Gamma}_r^\top(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{\Gamma}_r\mathbf{\Lambda}_r^{1/2}\mathbf{Z}$. The Moore–Penrose inverse is $\Sigma^+ = \mathbf{\Gamma}_r\mathbf{\Lambda}_r^{-1}\mathbf{\Gamma}_r^\top$, so $(\mathbf{Y} - \boldsymbol{\mu})^\top\Sigma^+(\mathbf{Y} - \boldsymbol{\mu}) = \|\mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top(\mathbf{Y} - \boldsymbol{\mu})\|^2 = \|\mathbf{Z}\|^2$, with expectation $\text{tr Cov}(\mathbf{Z}) = r$.

2.2.5.3 C. Going deeper

Exercise C1.

In $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ of full column rank, let $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{V}$ with $\mathbf{V}$ positive definite. Find $\text{Cov}(\hat{\boldsymbol{\beta}})$ for $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ and $\text{Cov}(\tilde{\boldsymbol{\beta}})$ for $\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{Y}$. Show that $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\tilde{\boldsymbol{\beta}})$ is nonnegative definite. *Hint:* show that $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}$ and $\tilde{\boldsymbol{\beta}}$ are uncorrelated.

Solution. By Theorem 2.2.1,

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{V}\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}, \quad \text{Cov}(\tilde{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})^{-1}.$$

Write $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}} = \mathbf{D}\mathbf{Y}$ with $\mathbf{D} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top - (\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{V}^{-1}$, and note $\mathbf{D}\mathbf{X} = \mathbf{I} - \mathbf{I} = \mathbf{0}$. Then $\text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\beta}}) = \sigma^2\mathbf{D}\mathbf{V}\mathbf{V}^{-1}\mathbf{X}(\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})^{-1} = \mathbf{0}$, so by Proposition 2.2.2(b), $\text{Cov}(\hat{\boldsymbol{\beta}}) = \text{Cov}(\tilde{\boldsymbol{\beta}}) + \text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})$, and the last term is nonnegative definite.